

Time Dilation and Length Contraction Near Light Speed

SimuPhysics Special Relativity Simulation · SimuPhysics

Senior Secondary (Years 11-12)	~30 minutes	Science · Physics
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Simulation: <https://simuphysics.com/view/modern-physics-special-relativity-simulation.html>
Curriculum links: special relativity · quantum physics

Learning intentions

- I can describe what happens to time and length as an object's speed approaches the speed of light.
- I can explain why relativistic effects are unnoticeable at everyday speeds.
- I can use simulation data to describe the relationship between velocity and the size of relativistic effects.

Getting started

Open the SimuPhysics Special Relativity Simulation. Start with the velocity slider at a low value (well below light speed) and gradually increase it. Watch how the time dilation factor and length contraction factor displayed by the simulation change as velocity increases.

Tasks

1. Set the velocity to a low value (e.g. 10% of light speed). Record the time dilation and length contraction factors shown.

2. Record the time dilation and length contraction factors at five different velocities, from low to very close to the speed of light.

Velocity (% of c)	Time dilation factor	Length contraction factor

3. Describe the pattern you see in your table as velocity increases toward the speed of light. Is the change in these factors steady, or does it speed up?

4. At everyday speeds (like a car or plane, well under 1% of light speed), would time dilation be something you could notice in daily life? Use your table to justify your answer.

5. Explain, using your data, why special relativity effects were not discovered through everyday observation, and instead required either very precise instruments or very high-speed particles to detect.

6. If a spacecraft could travel at 90% of the speed of light, and an astronaut on board experiences 1 year passing, use your simulation data at that velocity to estimate roughly how much time would pass for an observer on Earth.

Working:

7. Explain the relationship between time dilation and length contraction — do they both become significant at the same velocities, based on your data?

8. What happens to the time dilation factor as velocity approaches exactly the speed of light (100% of c)? What does this suggest about whether an object with mass could ever reach light speed?

Extension (optional)

GPS satellites orbit Earth at speeds high enough (and in weaker gravity) that relativistic effects must be corrected for, or their location data would drift by kilometres each day. Explain why even a small time dilation factor could cause this large a real-world error, given how fast GPS calculations need to be.
